

Safety data sheet

according to 1907/2006/EC, Article 31

Printing date 05.05.2021

Version number 22

Revision: 11.11.2020

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name: Böhler X 90-IG

CAS Number: -

EINECS Number: -

1.2 Relevant identified uses of the substance or mixture and uses advised against

No further relevant information available.

Application of the substance / the mixture

Rods and Wires for Welding

The product is a manufactured article in the sense of Article 3 No. 3, 1907/2006/EC (REACH). The purpose of the present safety data sheet is therefore to provide instruction on safe usage of the product.

1.3 Details of the supplier of the safety data sheet

Manufacturer/Supplier:

voestalpine Böhler Welding Germany GmbH
Hafenstr. 21
59067 Hamm, Germany
www.voestalpine.com/welding

Further information obtainable from:

Research and Development
Helena Stabel
+49 2381 271 - 578;
Helena.Stabel@voestalpine.com

1.4 Emergency telephone number:

NCEC

+44 1235 239670

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SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008

The Product does not meet the criteria for classification in any hazard class according to Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures.

2.2 Label elements

Labelling according to Regulation (EC) No 1272/2008 Void

Hazard pictograms Void

Signal word Void

Hazard statements Void

2.3 Other hazards

Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

SECTION 3: Composition/information on ingredients

3.2 Chemical characterisation: Mixtures

Description: Mixture of substances listed below with nonhazardous additions.

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Dangerous components:

CAS: 7440-02-0 EINECS: 231-111-4 Index number: 028-002-00-7 Reg.nr.: 01-2119438727-29-XXXX	nickel  Carc. 2, H351; STOT RE 1, H372  Skin Sens. 1, H317	0.1-2.5%
CAS: 7439-96-5 EINECS: 231-105-1 Reg.nr.: 01-2119449803-34-XXXX	manganese substance with a Community workplace exposure limit	0.1-2.5%

· **Additional information:** For the wording of the listed hazard phrases refer to section 16.

SECTION 4: First aid measures

· **4.2 Most important symptoms and effects, both acute and delayed**

No further relevant information available.

· **General information:** No special measures required.

· **After inhalation:** Supply fresh air; consult doctor in case of complaints.

· **After skin contact:** Generally the product does not irritate the skin.

· **After eye contact:** Rinse opened eye for several minutes under running water.

· **After swallowing:** Seek medical treatment.

· **4.3 Indication of any immediate medical attention and special treatment needed**

No further relevant information available.

SECTION 5: Firefighting measures

· **5.1 Extinguishing media**

· **Suitable extinguishing agents:** Suitable to surrounding conditions.

· **5.2 Special hazards arising from the substance or mixture** No further relevant information available.

· **5.3 Advice for firefighters**

For deletion of fire just use dry powders. Don't use any water or halogenated containing extinguishing agents

· **Protective equipment:** No special measures required.

SECTION 6: Accidental release measures

· **6.1 Personal precautions, protective equipment and emergency procedures**

Ensure adequate ventilation

Use respiratory protective device against the effects of fumes/dust/aerosol.

· **6.2 Environmental precautions:** No special measures required.

· **6.3 Methods and material for containment and cleaning up:** Pick up mechanically.

· **6.4 Reference to other sections**

See Section 7 for information on safe handling.

See Section 8 for information on personal protection equipment.

See Section 13 for disposal information.

SECTION 7: Handling and storage

· **7.1 Precautions for safe handling** Ensure that suitable extractors are available on processing machines

· **Information about fire - and explosion protection:** No special measures required.

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- **7.2 Conditions for safe storage, including any incompatibilities**
- **Storage:**
- **Requirements to be met by storerooms and receptacles:** No special requirements.
- **Information about storage in one common storage facility:** Not required.
- **Further information about storage conditions:** None.
- **7.3 Specific end use(s)** No further relevant information available.

SECTION 8: Exposure controls/personal protection

· 8.1 Control parameters

· **Ingredients with limit values that require monitoring at the workplace:**

7439-96-5 manganese

IOELV	Long-term value: 0.2* 0.05** mg/m ³ as Mn; *inhalable, **respirable fraction
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- **Additional information:** The lists valid during the making were used as basis.

· 8.2 Exposure controls

· **Personal protective equipment:**

- **General protective and hygienic measures:** Wash hands before breaks and at the end of work.

· **Respiratory protection:** Filter P2

· **Protection of hands:**

EN 12477

Selection of the glove material on consideration of the penetration times, rates of diffusion and the degradation

· **Material of gloves** Leather gloves

· **Penetration time of glove material**

The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.

· **Eye protection:** Safety glasses

· **Body protection:** Protective work clothing

SECTION 9: Physical and chemical properties

· 9.1 Information on basic physical and chemical properties

· **General Information**

· **Appearance:**

Form:

Solid

Colour:

Not determined.

· **Odour:**

Odourless

· **Odour threshold:**

Not determined.

· **pH-value:**

Not applicable.

· **Flash point:**

Not applicable.

· **Flammability (solid, gas):**

Not determined.

· **Decomposition temperature:**

Not determined.

· **Auto-ignition temperature:**

Product is not selfigniting.

· **Explosive properties:**

Product does not present an explosion hazard.

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· Explosion limits:	
Lower:	Not determined.
Upper:	Not determined.
· Density:	Not determined.
· Relative density	Not determined.
· Vapour density	Not applicable.
· Evaporation rate	Not applicable.
· water:	Insoluble.
· Partition coefficient: n-octanol/water:	Not determined.
· Dynamic:	Not applicable.
· Kinematic:	Not applicable.
· Solvent separation test:	
Solids content:	100.0 %
· 9.2 Other information	No further relevant information available.

SECTION 10: Stability and reactivity

- **10.1 Reactivity** No further relevant information available.
- **10.2 Chemical stability**
- **Thermal decomposition / conditions to be avoided:**
No decomposition if used and stored according to specifications.
- **10.3 Possibility of hazardous reactions** No dangerous reactions known.
- **10.4 Conditions to avoid** No further relevant information available.
- **10.5 Incompatible materials:** No further relevant information available.
- **10.6 Hazardous decomposition products:** No dangerous decomposition products known.

SECTION 11: Toxicological information

- **11.1 Information on toxicological effects**
- **Acute toxicity** Based on available data, the classification criteria are not met.
- **Primary irritant effect:**
- **Skin corrosion/irritation** Based on available data, the classification criteria are not met.
- **Serious eye damage/irritation** Based on available data, the classification criteria are not met.
- **Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.
- **Additional toxicological information:**
- **Repeated dose toxicity**
- **Germ cell mutagenicity** Based on available data, the classification criteria are not met.
- **Carcinogenicity** Based on available data, the classification criteria are not met.
- **Reproductive toxicity** Based on available data, the classification criteria are not met.
- **STOT-single exposure** Based on available data, the classification criteria are not met.
- **STOT-repeated exposure** Based on available data, the classification criteria are not met.
- **Aspiration hazard** Based on available data, the classification criteria are not met.

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SECTION 12: Ecological information

- **12.1 Toxicity**
- **Aquatic toxicity:** No further relevant information available.
- **12.2 Persistence and degradability** No further relevant information available.
- **12.3 Bioaccumulative potential** No further relevant information available.
- **12.4 Mobility in soil** No further relevant information available.
- **Additional ecological information:**
- **General notes:** Not hazardous for water.
- **12.5 Results of PBT and vPvB assessment**
- **PBT:** Not applicable.
- **vPvB:** Not applicable.
- **12.6 Other adverse effects** No further relevant information available.

SECTION 13: Disposal considerations

- **13.1 Waste treatment methods**
- **Recommendation** Must be specially treated adhering to official regulations.
- **European waste catalogue**
- 12 01 13 welding wastes
- **Uncleaned packaging:**
- **Recommendation:** Disposal must be made according to official regulations.

SECTION 14: Transport information

- | | |
|--|--|
| · 14.1 UN-Number | Void |
| · ADR, ADN, IMDG, IATA | Void |
| · 14.2 UN proper shipping name | |
| · ADR, ADN, IMDG, IATA | Void |
| · 14.3 Transport hazard class(es) | |
| · ADR, ADN, IMDG, IATA | |
| · Class | Void |
| · 14.4 Packing group | |
| · ADR, IMDG, IATA | Void |
| · 14.5 Environmental hazards: | |
| · Marine pollutant: | No |
| · 14.6 Special precautions for user | Not applicable. |
| · 14.7 Transport in bulk according to Annex II of Marpol and the IBC Code | Not applicable. |
| · Transport/Additional information: | Not dangerous according to the above specifications. |
| · UN "Model Regulation": | - |
| | Void |

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SECTION 15: Regulatory information

- **15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture**

No further relevant information available.

- **Directive 2012/18/EU**

- **Named dangerous substances - ANNEX I** None of the ingredients is listed.

- **REGULATION (EC) No 1907/2006 ANNEX XVII** Conditions of restriction: 27

- **DIRECTIVE 2011/65/EU on the restriction of the use of certain hazardous substances in electrical and electronic equipment – Annex II**

None of the ingredients is listed.

- **REGULATION (EU) 2019/1148**

- **Annex I - RESTRICTED EXPLOSIVES PRECURSORS (Upper limit value for the purpose of licensing under Article 5(3))**

None of the ingredients is listed.

- **Annex II - REPORTABLE EXPLOSIVES PRECURSORS**

None of the ingredients is listed.

- **15.2 Chemical safety assessment:** A Chemical Safety Assessment has not been carried out.

SECTION 16: Other information

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

- **Additional information:**

Recommendations for exposure scenarios, measures for risk management and identification of working conditions under which metals, metal alloys and products made of metal can be safely worked can be found attached.

Detailed information can be found on our webpage www.voestalpine.com (Environment, REACH at voestalpine).

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Welding Exposure Scenario WES - ENGL

EWA2011

Recommendations for Exposure Scenarios, Risk Management Measures and to identify Operational Conditions under which metals, alloys and metallic articles may be safely welded

Welding/Brazing produces fumes which can affect human health and the environment. Fumes are a varying mixture of airborne gases and fine particles which, if inhaled or swallowed, constitute a health hazard. The degree of risk will depend on the composition of the fume, concentration of the fume and duration of exposure. The fume composition is dependent upon the material being worked, the process and consumables being used, coatings on the work such as paint, galvanizing or plating, oil or contaminants from cleaning and degreasing activities. A systematic approach to the assessment of exposure is necessary, taking into account the particular circumstances for the operator and ancillary worker that can be exposed.

Considering the emission of fumes when welding, brazing or cutting of metals, it is recommended to (1) arrange risk management measures through applying general information and guidelines provided by this exposure scenario and (2) using the information provided by the Safety Data Sheet, issued in accordance with REACH, by the welding consumable manufacturer.

The employer shall ensure that the risk from welding fumes to the safety and health of workers is eliminated or reduced to a minimum. The following principle shall be applied:

- 1- Select the applicable process/material combinations with the lowest class, whenever possible.
- 2- Set welding process with the lowest emission parameter.
- 3- Apply the relevant collective protective measure in accordance with class number. In general, the use of PPE is taken into account after all other measures is applied.
- 4- Wear the relevant personal protective equipment in accordance with the duty cycle.

In addition, compliance with the National Regulations regarding the exposure to welding fumes of welders and related personnel shall be verified.

In the table "Risk Management Measures for individual process / material combinations" below, reference is made to the following standards for collective and personal protection measures:

ISO 4063	Welding process Reference Numbers according to ISO 4063
EN ISO 15012-1:2004	Health and safety in welding and allied processes - Requirements testing and marking of equipment or air filtration - Part 1: Testing of the separation efficiency for welding fume
EN ISO 15012-2:2008	Health and safety in welding and allied processes - Requirements, testing and marking of equipment for air filtration - Part 2: Determination of the minimum air volume flow rate of captor hoods and nozzles
EN 149:2001	Respiratory protective devices - Filtering half masks to protect against particles - Requirements, testing, marking (FFP1 - FFP2 - FFP3)
EN 1835:2000	Respiratory protective devices. Light duty construction compressed air line breathing apparatus incorporating a helmet or a hood. Requirements, testing, marking (LDH1 - LDH2 - LDH3).
EN 12941:1998	Respiratory protective devices. Powered filtering devices incorporating a helmet or a hood. Requirements, testing, marking (TH1 - TH2 - TH3).
EN 143:2000	Respiratory protective devices — Particle filters — Requirements, testing, marking (P1, P2, P3)
Directive 1998/24/EC	Article 6.2 on the protection of the health and safety of workers from the risks related to chemical agents at work
BGR 190	Benutzung von Atemschutzgeräten (Berufsgenossenschaftliche Regel für Sicherheit und Gesundheit bei der Arbeit)
TRGS 528	Schweisstechnische Arbeiten (Technische Regeln für Gefahrstoffe)

Also in the table "Risk Management Measures for individual process / material combinations", reference is made to footnotes.

The description of these footnotes:

- ¹ Class: approximate ranking to mitigate risk by selecting process/material combinations with the lowest value.
- ² Identified collective and individual risk management measures shall be applied
- ³ Personal Protective Equipment (PPE) required avoiding exceeding the National Exposure Limit Value (DC: Duty cycle expressed on 8 hours)
- ⁴ General Ventilation (GV) Low: With additional Local Exhaust Ventilation (LEV) and extracted air to the outside, the GV or LEV capacity may be reduced to 1/5 of the original requirement.
- ⁵ General Ventilation (GV) Medium (double compared to Low)
- ⁶ Filtrating half mask (FFP2)
- ⁷ When an alloyed consumable is used, measures from "Class V" are required
- ⁸ General Ventilation (GV) Low: When no Local Exhaust Ventilation, the ventilation requirement is 5-fold
- ⁹ Filtrating half mask (FFP3), helmet with powered filters (TH2/P2), or helmet with external air supply (LDH2)
- ⁹ Reduced (negative) pressured Area: A separate, ventilated area where reduced (negative) pressure, compared to the surrounded area, is maintained
- ¹⁰ Local Exhaust Ventilation (LEV) High, extraction at source (includes table, hood, arm or torch extraction)
- ¹¹ Helmet with powered filters (TH3/P3), or helmet with external air supply (LDH3)
- ¹² Local Exhaust Ventilation (LEV) Low, extraction at source (includes table, hood, arm or torch extraction)
- ¹³ Local Exhaust Ventilation (LEV) Medium, extraction at source (includes table, hood, arm or torch extraction)
- ¹⁴ Recommended measures to comply with national maximum allowable limits. Extracted fumes, for all materials except unalloyed steel and aluminium, shall be filtered before release in the outside environment.
- ¹⁵ A confined space, despite its name, is not necessarily small. Examples of confined spaces include ship, silos, vats, utility vaults, tanks, etc.
- ¹⁶ Improved helmet, designed to avoid direct flow of welding fumes inside
- ^{n.a.} Not applicable
- ^{n.r.} Not recommended

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Welding Exposure Scenario WES - ENGL

EWA2011

Risk Management Measures for individual process / base material combinations

Class ³	Process (according to ISO 4063)	Base Materials	Remarks	Ventilation / Extraction / Filtration ¹⁴	PPE ² DC<15%	PPE ² DC>15%		
Non-confined space ¹⁸								
I	GTAW 141	All	Except Aluminium	GV low ³	n.r.	n.r.		
	SAW 12							
	Autogeneous 3							
	PAW 15							
	ESW/EGW 72/73							
	Resistance 2							
	Stud welding 78							
	Solid state 521							
	Gases Brazing 9	All	Except Cd- alloys	GV low ³	n.r.	n.r.		
II	GTAW 141	Aluminium	n.a.	GV medium ⁴	n.a.	FFP2 ⁵		
III	MMAW 111	All	Except Be-, V-, Mn-, Ni- alloys and Stainless ⁵	GV low ⁷ LEV low ¹²	Improved helmet ¹⁶	FFP2 ⁵		
	FCAW 136/137	All	Except Stainless and Ni- alloys ⁶					
	GMAW 131/135	All	Except Cu-, Be-, V- alloys ⁵					
	Powder Plasma Arc 152	All	Except Be-, V-, Cu-, Mn-, Ni- alloys and Stainless ⁶					
IV	All processes class I	Painted / primed / oiled	No Pb containing primer	GV low ³	FFP2 ⁵	FFP3, TH2/P2, or LDH2 ¹⁷		
	All processes class III	Painted / primed / oiled	No Pb containing primer	GV low ⁴ / LEV low ¹²				
V	MMAW 111	Stainless, Ni-, Be-, and V- alloys	n.a.	LEV high ¹⁰	TH3/P3, LDH3 ¹¹	TH3/P3, LDH3 ¹¹		
	FCAW 136/137	Stainless, Mn- and Ni- alloys						
	GMAW 131	Cu- alloys						
	Powder Plasma Arc 152	Stainless, Mn-, Ni-, and Cu- alloys						
VI	GMAW 131	Be-, and V- alloys	n.a.	Reduced (negative) pressured area ⁹ LEV low ¹²	TH3/P3, LDH3 ¹¹	TH3/P3, LDH3 ¹¹		
	Powder Plasma Arc 152							
VII	Self shielded FCAW 114	Un-, high alloyed steel	Cored wire, not containing Ba	Reduced (negative) pressured area ⁹ LEV medium ¹³	TH3/P3, LDH3 ¹¹	TH3/P3, LDH3 ¹¹		
	Self shielded FCAW 114	Un-, high alloyed steel	Cored wire, containing Ba	Reduced (negative) pressured area ⁹ LEV high ¹⁰				
	All	Painted / primed	Paint / Primer containing Pb					
	Arc Gouging and Cutting 8	All	n.a.					
	Thermal Spray	All	n.a.					
	Gases Brazing 9	Cd- alloys	n.a.					
Closed system or Confined space ¹⁸								
I	Laser Welding 52	All	Closed system	GV medium ⁴	n.a.	n.a.		
	Laser Cutting 84							
	Electron Beam 51							
VIII	All	All	Confined space	LEV high ¹⁰ External air supply	LDH3 ¹¹	LDH3 ¹¹		

Relevant phrases

H317 May cause an allergic skin reaction.

H351 Suspected of causing cancer.

H372 Causes damage to organs through prolonged or repeated exposure.

Department issuing SDS: R&D

Contact: Helena Stabel

Abbreviations and acronyms:

NCEC - National Chemical Emergency Centre (=Carechem24)

ADR: Accord relatif au transport international des marchandises dangereuses par route (European Agreement Concerning the International Carriage of Dangerous Goods by Road)

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IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

GHS: Globally Harmonised System of Classification and Labelling of Chemicals

EINECS: European Inventory of Existing Commercial Chemical Substances

ELINCS: European List of Notified Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

TRGS: Technische Regeln für Gefahrstoffe (Technical Rules for Dangerous Substances, BAuA, Germany)

PBT: Persistent, Bioaccumulative and Toxic

vPvB: very Persistent and very Bioaccumulative

Skin Sens. 1: Skin sensitisation – Category 1

Carc. 2: Carcinogenicity – Category 2

STOT RE 1: Specific target organ toxicity (repeated exposure) – Category 1

*** Data compared to the previous version altered.**

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